

Dual Reasoning: A GNN-LLM Collaborative Framework for Knowledge Graph Question Answering

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Large Language Models (LLMs) excel at intuitive, implicit reasoning. Guiding LLMs to construct thought chains can enhance their deliberate reasoning abilities, but also faces challenges such as hallucination. Knowledge Graphs (KGs) can provide explicit structured knowledge for LLMs to alleviate these issues. However, existing KG-enhanced methods often overlook explicit graph learning, making it challenging to efficiently provide precise reasoning chains for LLMs. Following dual-process theory, we propose Dual-Reasoning (DualR), a novel framework that integrates an external system based on Graph Neural Network (GNN) for explicit reasoning on KGs, complementing the implicit reasoning of LLMs through externalized reasoning chains. DualR designs an LLM-empowered GNN module for explicit learning on KGs, efficiently extracting high-quality reasoning chains. These reasoning chains are then refined to a knowledge-enhanced multiple-choice prompt, guiding a frozen LLM to reason thoughtfully for final answer determination. Extensive experiments on three benchmark KGQA datasets demonstrate that DualR achieves state-of-the-art performance while maintaining high efficiency and interpretability. Our code and data are available in <https://github.com/leolouis14/DualR>.

1. Introduction

Large language models (LLMs) [1–4] have demonstrated impressive capabilities across various natural language processing tasks. Pre-trained on extensive corpora, LLMs excel at implicit and associative reasoning. To further enhance LLMs’ reasoning abilities, many approaches (e.g., Chain-of-Thought (CoT) [5], Tree-of-Thought (ToT) [6]) guide LLMs to generate intermediate steps and form a complete thought chain, aiming for a more deliberate and explicit reasoning [7, 8]. While these approaches can improve performance, they typically increase inference costs, and encounter the challenge of hallucinations when the model lacks relevant knowledge, especially domain-specific and up-to-date knowledge [7, 9].

Integrating external knowledge sources, such as knowledge graphs (KGs), offers a promising solution to these limitations. KGs, storing a vast amount of facts in the form of triples (e.g., Wikidata [10], YAGO [11], and NELL [12]), are vital for a variety of applications due to their capacity to deliver explicit knowledge [13, 14]. A common and essential task for integrating LLMs with KGs is Question Answering over Knowledge Graph (KGQA), which aims at answering natural language question from entities within a given KG. To accurately respond to a given question, a key challenge is that how to enable LLMs to effectively acquire supportive reasoning evidence from a large and complex KG structure. Existing methods frequently overlook the importance of explicit learning within graph structures, which is essential for supplying precise evidence chain for LLM’s reasoning. The text-based retrieval methods (e.g., KAPING [15]), directly retrieve triplets based on text similarity from KGs for LLMs, which frequently result in redundant or irrelevant information [16]. Another LLM-based retrieval methods like StructGPT [17], ToG [18], guide LLMs to retrieve over KGs across multiple steps. Since LLMs lack the inherent capacity to comprehend graph structures, it is often challenging for them to perform effective topological reasoning on graphs [9]. Furthermore, frequent

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interaction with LLMs entails significant time and resource costs, especially in the KGQA task that requires multi-hop reasoning on a large KG.

To effectively integrate LLMs with KGs and address challenges above, we draw on cognitive science for inspiration. Bengio et al. [19] has highlighted connections between deep learning and the cognitive process described in Daniel Kahneman’s *Thinking fast and Slow* [20]. The dual-process theory in the book posits that human thinking arises from two complementary systems: the intuitive and implicit System1, and the deliberate and explicit System 2 [21, 22]. Building on this, we introduce Dual-Reasoning (DualR), a novel approach that incorporates an external, structured “System 2” to explicitly reason on KGs, extracting valuable reasoning chains to complement the intuitive and implicit “System 1”-like reasoning of LLMs. To efficiently implement this externalized “System 2”, we utilize graph neural networks (GNNs) [23–25], which are well-suited for learning within complex graph structures. Specifically, we first design an LLM-empowered GNN module to explicitly reason on the KG, extracting high-quality reasoning chains relevant to the question. Then, the identified reasoning chains are refined into a knowledge-enhanced multiple-choice prompt, guiding a frozen LLM to reason thoughtfully for final answer determination. With the collaboration of “System 1” and “System 2”, DualR enables efficient and explicit reasoning over KGs, enhancing the reasoning capabilities of LLMs through extracted reasoning chains, and achieving accurate and faithful results in KGQA task. Furthermore, the framework is designed to integrate seamlessly with any off-the-shelf LLMs with just one-step inference, without requiring extensive interactions or fine-tuning, thus allowing for resource-efficient deployment. The contributions are summarized as follows:

- Following dual-process theory, we introduce a novel framework, Dual-Reasoning (DualR), which integrates an external, structured “System 2” for deliberate, explicit reasoning on KGs, complementing the implicit reasoning of LLMs through externalized reasoning chains.
- To implement this framework, we design a lightweight GNN model empowered by LLM for precise and efficient reasoning on KG, extracting high-quality reasoning chains. We further propose a knowledge-enhanced multiple-choice prompt to guide the LLM to reason for final answer.
- Extensive experiments on KGQA show that our method effectively combines GNN’s structured, explicit learning with LLM’s powerful language understanding, outperforming state-of-the-art methods while maintaining high efficiency and interpretability.

2. Related Work

Dual-Process Theory. Dual-process theory [20–22] is a psychological account of how human thinking and decision-making arise from two distinct systems. System 1, corresponding to the implicit process, is associative and intuitive, enabling quick comprehension through associations and pre-existing knowledge [7, 8]. System 2, corresponding to the explicit process, is more deliberate and logical. It operates on symbolic structures, conducting explicit reasoning to arrive at conclusions [8, 22]. These systems serve complementary functions and can collaborate for a reasoning problem [22].

LLM reasoning. LLMs [1–4], have demonstrated impressive capabilities in many tasks such as question answering by leveraging implicit, associative reasoning similar to the intuitive “System 1” process in dual-process theory [20, 22]. To further improve their performance, some approaches instruct LLMs to generate reasoning process in their outputs [5, 6, 26]. While these “System 2”-like deliberate and explicit reasoning methods can achieve better results, they typically entail higher inference costs [7]. Furthermore, LLMs may make mistakes during the reasoning process, due to limitations in their internal knowledge, leading to issues such as hallucinations and inaccuracies [9].

KGQA. Given a natural language question q and a KG $\mathcal{G} = \{(e_s, r, e_o) | e_s, e_o \in \mathcal{V}, r \in \mathcal{R}\}$, where $\mathcal{V}$ is the set of entities (nodes) and $\mathcal{R}$ is the set of relation types, the task of *Question Answering over Knowledge Graph* (KGQA) is to find a function $\mathcal{F}(q, \mathcal{G})$ that predicts the answer entities $e_a \in \mathcal{V}$ of q over KG $\mathcal{G}$. As a common and practical setting, for each question q , the involved topic entities $e_q \in \mathcal{V}$ and answer entities $e_a \in \mathcal{V}$ are both labeled in KG. KGQA plays a vital role in various intelligent systems, such as Apple Siri and Microsoft Cortana [27]. To solve this task, classical methods [28–31]

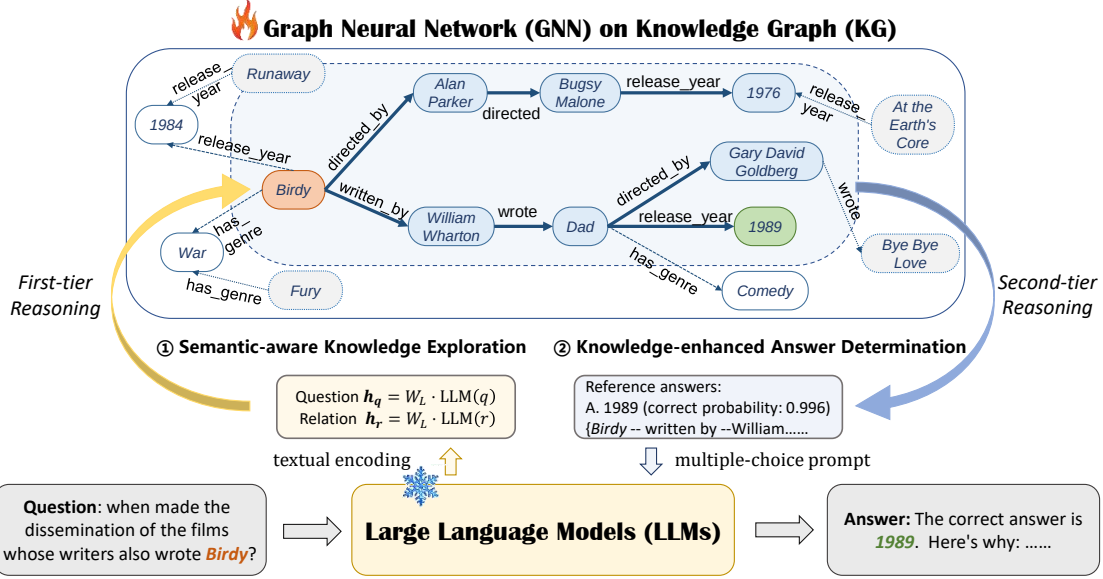


Figure 1: Illustration of the Dual-Reasoning (DualR), a GNN-LLM collaborative framework for knowledge graph question answering. It contains two-tier reasoning: (1) semantic-aware knowledge exploration; and (2) knowledge-enhanced answer determination.

usually first retrieve a question-related subgraph and then use different model (e.g., embedding model, graph neural networks[23, 24]) to reason for answers.

LLM for KGQA. Recently, considering the powerful language processing capabilities of LLMs, many works integrate LLMs with KG for KGQA task. These works typically follow the RAG paradigm [32, 33], which utilize various modules to extract knowledge from KG for LLMs generating final answers. Some methods retrieve information from KG based on text similarity [15, 34, 35]. For instance, KAPING [15] employs embedding model to retrieve triplets, but fails to leverage structured knowledge to do reasoning, resulting in the retrieval of excessive and irrelevant information [36]. Another strategy involves guiding the LLM itself to retrieve the knowledge from KG [17, 18, 37, 38]. For example, StructGPT [17] and ToG [18] view LLM as an agent, guiding it to search on the KG iteratively. But the depth and breadth of this search are limited as LLMs inherently lack the ability to comprehend graph structures [9]. Additionally, frequent interactions with LLMs are inflexible and entail high costs. RoG [38] fine-tunes an LLM to generate reasoning paths for information retrieval, but it requires substantial fine-tuning and may generate invalid path due to the hallucinations. A concurrent approach, GNN-RAG [39], utilizes an off-the-shelf GNN model to retrieve the shortest path to high-scoring candidate entities for LLMs. However, it depends on heuristic technique for path retrieval, which may cause low-quality reasoning chains. Moreover, supplying only the path does not fully leverage the GNN model, limiting its effectiveness. Notably, there is an independent category of methods, semantic parsing methods, which use LLMs to generate query languages and then execute them on KGs to obtain answers [40–43]. These methods typically rely on SPARQL query and are orthogonal to ours.

3. Proposed Method

3.1. Overview

Constructing high-quality reasoning chains is crucial for LLMs to generate correct answers. However, in the KGQA task, LLMs face challenges in effectively and efficiently extracting meaningful reasoning chains relevant to a given question from large and complex KGs. Following dual-process theory that describes human thinking as arising from two complementary systems, we propose a novel Dual-Reasoning (DualR) framework, which combines the strengths of GNN-based “System 2” and LLM-based “System 1” to achieve accurate, efficient and faithful reasoning over KGs.

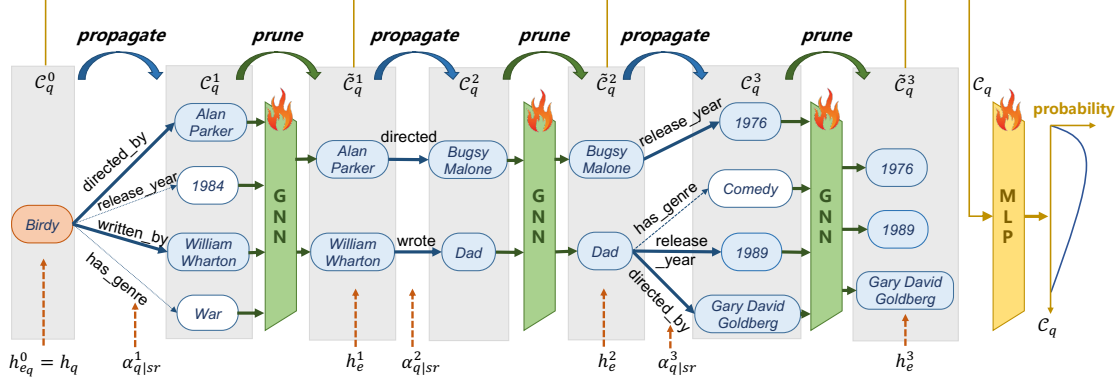


Figure 2: Illustration of semantic-aware knowledge exploration. We start exploration from topic entity *Birdy*. In each step, we firstly get the unpruned candidate set C_q^ℓ , calculate the attention weights $\alpha_{q|sr}^\ell$ of different edges, prune several irrelevant entities (in white), and update candidate set $\tilde{C}_q^\ell$ (in blue). The representations are propagated from entities in $\tilde{C}_q^{\ell-1}$ to $\tilde{C}_q^\ell$ through an one-layer GNN.

The proposed framework is shown in Figure 1, which contains two-tier reasoning: (1) semantic-aware knowledge exploration; and (2) knowledge-enhanced answer determination. For the first part, to implement the “System 2” efficiently, we design an LLM-empowered GNN module to explicitly reason on the graph, adaptively exploring fine-grained structured knowledge to construct meaningful reasoning chains. For the second part, to effectively leverage the extracted reasoning chains, we carefully design a knowledge-enhanced multiple-choice prompt, guiding the LLM to perform thoughtful reasoning, achieving accurate and reliable answer determination.

3.2. Semantic-Aware Knowledge Exploration

In this section, we aim to construct a deliberate, structured “System 2”-like module on the KG. Through explicit learning and reasoning, it explores structured knowledge on the graph, extracting meaningful reasoning chains that connect topic entity with promising candidates. Due to a large amount of irrelevant information present in the KG, we design an adaptive exploration GNN module that, with the semantic representation capabilities provided by the LLM, can automatically prune irrelevant information, enabling precise, efficient and interpretable reasoning on KG.

Semantic-aware pruning. Given the topic entity e_q involved in the question q , we aim to gradually explore the potential answer entities starting from e_q on the KG. Instead of directly using LLM to explore the candidates, we design a lightweight neural network empowered by LLM. Specifically, we initialize the candidate set $C_q^0 \equiv \{e_q\}$. The representation of e_q is initialized as the question encoding, i.e., $h_{e_q}^0 = h_q = W_L \cdot \text{LLM}(q)$, where $\text{LLM}(\cdot)$ computes the average embedding in the first and last layer of Llama2-13B[3], and $W_L \in \mathbb{R}^{d \times d_L}$ is a learnable weighting matrix mapping the representation to a lower dimension d . The representations of other entities are initialized as 0. Assume in the ℓ -th step ($\ell = 1, 2, \dots, L$), we have explored a set $C_q^{\ell-1}$ of current candidates. The set is then expanded by propagating to the neighbors of entities in $C_q^{\ell-1}$, resulting in an updated set $C_q^\ell = \{e_o : (e_s, r, e_o) \in \mathcal{G}, e_s \in C_q^{\ell-1}\}$.

Since KG contains a lot of information irrelevant to the question, we should filter the irrelevant edges during exploring such that the size of C_q^ℓ will not grow exponentially. In the ℓ -th step, we calculate the attention weight $\alpha_{q|sr}^\ell$ to measure the importance of each edge (e_s, r, e_o) with $e_s \in C_q^{\ell-1}$ as:

$$\alpha_{q|sr}^\ell = \sigma \left(W_s^\ell h_s^{\ell-1} + W_r^\ell h_r + W_q^\ell h_q + W_{qr}^\ell (h_r \odot h_q) \right), \quad (1)$$

where σ is the sigmoid function, W^ℓ 's in $\mathbb{R}^{1 \times d}$ are learnable weight matrices, and $\odot$ is the Hadamard product of vectors. The representations h_q and h_r are textual encodings mapped from an LLM (i.e., Llama2-13B) such that the semantic relevance of question with the current edge can be measured. Note that the LLM used as text encoding will not be updated. The representation $h_s^{\ell-1} \in \mathbb{R}^d$ of head entity e_s contains the knowledge learned in the $(\ell - 1)$ -th step.

By incorporating the representations of question q , relations, and entities within the triplets, we can utilize the powerful semantic modeling capabilities of LLM. Meanwhile, setting different weight matrices in each step allows us to capture essential information from different sections of the question, thereby preserving the triplets that are semantically relevant to the question. For each head node e_s , we select the top- K edges (e_s, r, e_o) based on $\alpha_{q|sr}^\ell$ of different edges and prune the others, resulting in a smaller candidate set $\tilde{\mathcal{C}}_q^\ell$. Here, K is a hyperparameter based on the characteristics of the KG. For example, Figure 2 illustrates a simple example with $K = 2$ and $L = 3$, where blue entities are retained while white entities irrelevant to the question are pruned. This way, we can adaptively explore the related information on the graph while reducing computation costs.

GNN encoding through propagation. To learn representations of each entity $e_o \in \tilde{\mathcal{C}}_q^\ell$, we use a lightweight network, i.e., a 1-layer GNN, to propagate the information from entities $e_s \in \tilde{\mathcal{C}}_q^{\ell-1}$ one step further to entities e_o with

$$\mathbf{h}_o^\ell = \delta\left(\sum_{(e_s, r, e_o) \in \tilde{\mathcal{N}}_{e_o}^\ell} \alpha_{q|sr}^\ell \mathbf{W}^\ell(\mathbf{h}_s^{\ell-1} \odot \mathbf{h}_r)\right), \quad (2)$$

where $\delta(\cdot)$ is the activation function, $\tilde{\mathcal{N}}_{e_o}^\ell$ is the set of preserved neighbor edges of tail entity e_o , $\mathbf{W}^\ell \in \mathbb{R}^{d \times d}$ is a learnable weight matrix in the ℓ -th step, and the attention weight $\alpha_{q|sr}^\ell$ is computed in (1). In this way, the compositional information of edges relevant to question q connecting from topic entity e_q to e_o can be propagated into $\mathbf{h}_o^\ell$.

As shown in Figure 2, after L steps of propagation, we can form the final candidate set $\mathcal{C}_q = \tilde{\mathcal{C}}_q^0 \cup \dots \cup \tilde{\mathcal{C}}_q^L$ and obtain their representations $\mathbf{h}_e^L$. Finally, we use a multi-layer perceptron (MLP) and softmax function on the entity representation $\mathbf{h}_{e_i}^L$ and the question representation $\mathbf{h}_q$ to obtain the probability of entity e_i being the correct answer:

$$p(q, e_i) = e^{\text{MLP}([\mathbf{h}_{e_i}^L; \mathbf{h}_q])} / \sum_{e_j \in \mathcal{C}_q} e^{\text{MLP}([\mathbf{h}_{e_j}^L; \mathbf{h}_q])}. \quad (3)$$

We optimize the neural network with supervision given by the question-answer pairs. Specifically, we use the CrossEntropy Loss [44]:

$$\mathcal{L} = \sum_{(q, e_a) \in \mathcal{F}_{tra}} -\log(p(q, e_a)), \quad (4)$$

where $\mathcal{F}_{tra}$ is the training set of question-answer pairs. The set of model parameters are randomly initialized and optimized by minimizing $\mathcal{L}$ with Adam stochastic gradient descent algorithm [45].

Through adaptive propagation with GNN, the ‘‘System 2’’-like explicit reasoning we build on the KG can effectively explore structured knowledge, identifying promising candidate answers. Furthermore, by leveraging the weights of different edges, we can extract paths connecting the topic entity and candidate answers to construct meaningful reasoning chains, which will be detailed in Section 3.3.

3.3. Knowledge-Enhanced Answer Determination

Although the GNN module can indicate the probabilities of candidates from the output scores, leveraging the powerful language understanding capabilities of the LLM for aligning the question linguistically with the explored information can provide greater benefits. Therefore, we extract high-quality reasoning chains to guide the LLM in reasoning for final answer determination. Specifically, to enhance the associative ‘‘System 1’’-like reasoning of LLMs, we design a knowledge-enhanced multiple-choice prompt, guiding LLMs in effectively combining external explicit knowledge with its own internal implicit knowledge, and generating answers with a rapid one-step inference.

Reasoning chain extraction. To extract valuable reasoning chains for LLM, we employ a greedy algorithm to trace back paths. Specifically, we first preserve top- N candidate entities in $\mathcal{C}_q$ identified with the highest probabilities as reference answers. Then, we backtrack from each candidate answer e_c , select the edge from $\tilde{\mathcal{N}}_{e_c}^L$ with the highest attention weight (as per (1)), and set the head entity as a new starting point in the next step. By conducting L -steps backtracking, it will eventually trace back to the initial topic entity e_q and obtain a reasoning evidence chain between e_q and e_c (the detailed

algorithm is shown in Appendix B.3). These chains reveal the compositional associations between topic entities and candidate answers, providing faithful evidence for the final answer determination. For example, in Figure 1, the evidence chain connecting the topic entity *Birdy* and the reference answer 1989 is: *Birdy* $\xrightarrow{\text{written_by}}$ William Wharton $\xrightarrow{\text{wrote}}$ Dad $\xrightarrow{\text{release_year}}$ 1989. In this way, LLMs can conduct semantic analysis and reasoning based on reasoning chains. For instance, it can recognize that the second reference answer (1976) in Figure 1 does not align with the semantic context of the question based on the chains between *Birdy* and 1976.

Knowledge-enhanced multiple-choice prompt. To effectively leverage the extracted reasoning chains, we further refine them into a knowledge-enhanced multiple-choice prompt. We incorporate two additional types of knowledge to enhance the LLM’s reasoning for answer determination. First, we extract the list of candidate entity and assign them labels. This approach provides clear targets, effectively stimulating the LLM’s associative and intuitive reasoning capabilities. Second, we use the correct probability (as per (3)) of each candidate answer to provide the confidence returned by the GNN module for LLM, which aids the LLM in engaging in reasoning with heightened attention.

Overall, the knowledge-enhanced multiple-choice prompt is built using a task description, the provided question, and reference answers, which include candidate entity, correct probability and reasoning chain, as shown on the right.

```
<Task Description>
<Question>
<Reference Answers>:
A. candidate 1 (correct probability) {reasoning chain}
B. candidate 2 (correct probability) {reasoning chain} .....
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The complete input-output examples of the final answer determination are provided in the Appendix D, which demonstrates our method’s capability for accurate and interpretable reasoning. In addition, to avoid distractions from irrelevant information, we do not offer any few-shot examples, providing a clear structure for the LLM to follow, while maintaining the benefits of zero-shot learning.

3.4. Training Strategy

During training, we freeze the LLM in both text encoding and answer determination steps to avoid expensive costs. Instead, we only update the lightweight GNN module by minimizing the loss in (4). Moreover, unlike traditional GNN-based approaches [29–31], which first retrieve a subgraph and then perform reasoning on it, our GNN module can unify and simultaneously perform these two steps, adaptively filter out irrelevant information on the graph, and achieve more efficient reasoning. Additionally, it is noteworthy that the answer determination module can be adapted to any pre-trained LLM with just one-step inference, without the need for fine-tuning or frequent interaction, thereby avoiding time and resource overhead. In this way, the GNN module can serve as “System 2”, as it performs explicit, step-by-step reasoning on graph structures, while the LLM aligns with “System 1”, leveraging implicit, associative patterns for rapid and intuitive decision-making.

Considering there are common compositional relationships between questions and the concepts in the graphs, the GNN module can be further benefited from pre-training strategies. Specifically, we pre-train the networks on two comprehensive KGQA datasets WebQSP [46] and CWQ [47], and then fine-tune them on target datasets. This approach enables the GNN module to better learn the compositional relationships, enhancing question understanding and generalization abilities.

4. Experiments

4.1. Experimental Setup

Datasets. Following existing KGQA works [30, 31], we use three benchmark datasets, namely WebQSP [46], CWQ [47], and MetaQA [48], to evaluate different methods. The MetaQA dataset is divided into three versions based on the number of hops required in KG, namely 1-hop, 2-hop, and 3-hop. Table 1 displays the statistics of these three datasets.

Table 1: Statistics of KGQA datasets.

Datasets	#Train	#Valid	#Test	Max #hop
WebQSP	2,848	250	1,639	2
CWQ	27,639	3,519	3,531	4
MetaQA-1	96,106	9,990	9,947	1
MetaQA-2	118,980	14,872	14,872	2
MetaQA-3	114,196	14,274	14,274	3

Table 2: Performance comparison of different methods for KGQA (Hits@1 in percent).

Type	Methods	WebQSP	CWQ	MetaQA-1	MetaQA-2	MetaQA-3
KG-based	KV-Mem	46.7	18.4	96.2	82.7	48.9
	GraftNet	66.4	36.8	97.0	94.8	77.7
	EmbedKGQA	66.6	-	97.5	98.8	94.8
	NSM	68.7	47.6	97.1	99.9	98.9
	SR+NSM	69.5	50.2	-	-	-
	UniKGQA	75.1	50.7	<u>97.5</u>	99.0	99.1
GPT-3-based	KB-Binder	74.4	-	92.9	99.9	<u>99.5</u>
	KAPING	73.9	55.4	-	-	-
Llama2-based	Llama2-13B	40.9	22.1	31.9	15.8	34.9
	RoG-Llama2-7B	74.2	56.4	-	-	-
	ToG-Llama2-70B	68.9	57.6	-	-	-
	DualR-Llama2-13B	78.3	58.0	97.9	99.1	99.6
ChatGPT-based	ChatGPT	61.2	38.8	61.9	31.0	43.2
	RoG-ChatGPT	81.5	52.7	-	-	-
	KD-CoT	68.6	55.7	-	-	-
	StructGPT	72.6	55.3	94.2	93.9	80.2
	ToG-ChatGPT	76.2	58.9	-	-	-
	DualR-ChatGPT	82.8	62.0	98.1	99.7	99.7
GPT-4-based	GPT-4	67.3	46.0	65.7	34.6	48.9
	ToG-GPT-4	82.6	69.5	-	-	-
	DualR-GPT-4	87.6	73.6	98.3	99.9	99.9

Evaluation Metrics. Following [17, 18], we focus on generating the answer with the highest confidence, and use Hits@1 to evaluate whether the top-ranked predicted answer is correct.

Experiment Details. In the pre-training stage, we set the dimension d as 256 for the GNN module, learning rate as $1e-4$, batch size as 20, number of layers L as 3 and number of sampling K as 200. As for the fine-tuning stage, we adjust the L and K based on the performance on validation set, and details are described in Appendix B.4. Considering the plug-and-play convenience of DualR, we use three LLMs for answer determination in experiments: Llama2-13B-chat [3], ChatGPT and GPT-4². We typically set number of reference answers N as 3, and the influence of N is shown in Appendix C.4. The GNN module is trained on an RTX 3090-24GB GPU, while inference for Llama2-13B-chat runs on two RTX 3090-24GB GPUs.

Baseline Methods. We consider following baseline methods for performance comparison: (1) KG-based methods without using LLMs: KV-Mem [49], GraftNet [29], EmbedKGQA [28], NSM [30], SR+NSM [50], UniKGQA [31]; (2) LLM-based methods: KB-Binder [42] based on Codex [51], KAPING [15] based on GPT-3 [1], RoG [38] that can be plug-and-play with different LLMs, KD-CoT [37] and StructGPT [17] based on ChatGPT, ToG [18] that can be plug-and-play with different LLMs.

4.2. Performance Comparison

Main results. From the results in Table 2, it can be observed that our method DualR, whether combined with the Llama2 or GPT, outperforms traditional methods without LLMs. From the last three groups of the table, it can be seen that incorporating KG can effectively enhance the performance of LLMs, as they often lack the knowledge relevant to the questions. Within methods using Llama2, DualR combined with the 13B model outperforms ToG which is combined with the 70B model, and also outperforms some methods using ChatGPT. Similarly, within methods using ChatGPT and GPT-4, our approach demonstrates significant advantages. This shows the superiority of proposed dual-reasoning framework, effectively harnessing the GNN’s precise explicit learning and the LLM’s powerful language understanding, outperforming LLM’s single reasoning significantly. Additionally, the performance of DualR improves with the integration of more powerful LLM. This demonstrates that DualR effectively combines explicit knowledge from KGs with implicit knowledge from LLM, leading to more accurate reasoning.

²<https://openai.com/>

Table 3: Comparison of inference time (seconds) and number of interaction with LLM (Llama2-13B) per question of different methods.

Methods	WebQSP		CWQ	
	time	#interaction	time	#interaction
RoG	1.98	2	3.04	2
StructGPT	3.37	3	4.22	4
ToG	16.7	15	20.5	22
DualR	1.29	1	1.99	1

Table 4: Comparison of different variants of DualR-Llama2-13B in Hits@1(%).

Methods	WebQSP	CWQ
DualR	78.3	58.0
w.o.-mcp	72.9	54.1
w.o.-cand	77.5	52.5
w.o.-prob	76.2	55.4
w.o.-chain	76.8	56.0

Efficiency comparison. We also compare the inference time and number of interaction with LLM of several representative LLM-based methods, executing Llama2-13B within our local environment. As can be seen in Table 3, RoG requires two interactions with LLM, while step-by-step reasoning methods StructGPT and ToG require even more frequent interactions, thus incurring a high cost. However, in our approach, question encoding of LLM and graph reasoning of GNN are highly efficient, and we ultimately requires only one-step inference of LLM, thereby achieving high efficiency.

Fine-tuned setting results. While our method can seamlessly integrate with any off-the-shelf LLM, we expect that instruction fine-tuning the LLM for answer determination can lead to further performance improvements. We compare our approach with two recent state-of-the-art methods, RoG and GNN-RAG [39], both of which fine-tune Llama2-7B-chat on WebQSP and CWQ datasets, using Hit, Hits@1 and F1 as evaluation metrics. As shown in Table 5, DualR achieves state-of-the-art performance across multiple metrics, especially showing greater gains on more challenging dataset (i.e., CWQ), further validating the effectiveness of our method.

Table 5: Performance comparison of different methods with fine-tuned Llama2-7B-chat.

Methods	WebQSP			CWQ		
	Hit	Hits@1	F1	Hit	Hits@1	F1
RoG	85.7	80.0	70.8	62.6	57.8	56.2
GNN-RAG	85.7	80.6	71.3	66.8	61.7	59.4
DualR	84.9	81.5	71.6	68.9	65.3	62.1

4.3. Ablation Study

4.3.1. Effectiveness of Knowledge Exploration

In this section, we analyze the effectiveness of our GNN module, focusing on the Hits@1 of its independent output answer without the answer determination by LLM (i.e., DualR-w.o.-AD). Here, we choose UniKGQA, the state-of-the-art KG-based model without LLMs for comparison. As can be seen in Table 6, our GNN module outperforms UniKGQA on all datasets, demonstrating the effectiveness of the designed network for topology and semantic aware reasoning on the graph, capable of identifying promising candidates and extracting high-quality reasoning chains.

4.3.2. Effectiveness of LLM’s Answer Determination

Furthermore, Table 6 demonstrates the effectiveness of using LLM to reason for answer determination. Although the GNN module itself (DualR-w.o.-AD) achieves decent performance, leveraging extracted reasoning chains to guide LLM for answer determination brings significant gains. With its powerful language processing capabilities, the LLM can analyze the semantic relationships between a given question and relevant reasoning chains and make the thoughtful determination. By leveraging the implicit knowledge embedded within itself, it achieves more accurate reasoning.

4.3.3. Influence of Prompt Form for Answer Determination

We design different forms of prompts to evaluate their influence in guiding answer determination of DualR-Llama2-13B. The variants include: (1) not using multiple-choice prompt (DualR-w.o.-mcp), (2) not using candidate entities (DualR-w.o.-cand), (3) not using correct probabilities (DualR-w.o.-prob), and (4) not using reasoning chains (DualR-w.o.-chain), which are shown in Appendix E.

Table 6: Comparison of Different Variants of DualR in Hits@1(%).

Methods	WebQSP	CWQ	MetaQA-1	MetaQA-2	MetaQA-3
UniKGQA	75.1	50.7	97.5	99.0	99.1
DualR-w.o.-AD	76.8	55.6	97.6	99.1	99.5
DualR-Llama2-13B	78.3	58.0	97.9	99.1	99.6
DualR-ChatGPT	82.8	62.0	98.1	99.7	99.7

As shown in Table 4, compared with DualR-w.o.-mcp, which inputs all candidate answers continuously without specific labels, our designed multiple-choice prompt can achieve better performance, since it offers a clearer structure and strength the connection of candidate answers with reasoning chains. Moreover, all three factors in the prompt, i.e., candidate entity, correct probability and reasoning chain, have a positive impact on guiding LLM to make correct choices. Providing candidate answers can refine the target for LLM (compared with DualR-w.o.-cand), and correct probabilities prompts LLM to engage in reasoning with attention (compared with DualR-w.o.-prob). Reasoning paths are crucial as they provide detailed process for LLM reasoning (compared with DualR-w.o.-chain), enhancing the accuracy and reliability of the output, as can be seen in Section 4.4.

4.4. Case Study

We present two case studies in Figure 3, which display the question, reference answers of the input prompt, and the final result of the LLM’s answer determination. The task description can be found in Appendix D. It can be observed that the LLM without Dual-Reasoning (i.e., Llama2-13B-CoT) may suffer from hallucination and lack of knowledge, resulting in wrong answers. In contrast, DualR-Llama2-13B generates the correct answers, even when the right answer is not ranked at the top. Moreover, based on the given information, the LLM provides its reasoning process, which is consist with the input reasoning chains, enhancing the credibility of the output. These examples illustrate how DualR effectively implement LLM’s intuitive reasoning through externalized reasoning chains, achieving accurate and faithful results.

Question-Answer	When made the dissemination of the films whose writers also wrote Birdy? - 1989.	Question-Answer	What is the most current year the team who won the 2010 World Series has won another World Series? - 2014 World Series.
Output by Llama2-13B-CoT	The films whose writers also wrote Birdy were disseminated in the following years: * The Right Stuff (1983) * The Big Red One (1980) Both of these films were written by Larry McMurtry and Daniel Taradash, who also co-wrote the screenplay for Birdy.	Output by Llama2-13B-CoT	The team that won the 2010 World Series, the San Francisco Giants, has not won another World Series since then. Therefore, the most current year they have won a World Series is 2010.
Input Question and Reference Answers	Question: when made the dissemination of the films whose writers also wrote Birdy? Reference answers: A. 1989 (correct probability: 0.996) {relevant facts: (Birdy, written_by, William Wharton), (Dad, written_by, William Wharton), (Dad, release_year, 1989)} B. 1998 (correct probability: 0.004) {relevant facts: (Birdy, has_tags, nicolas cage), (Snake Eyes, has_tags, nicolas cage), (Snake Eyes, release_year, 1998)} C. 1976 (correct probability: 0.0) {relevant facts: (Birdy, directed_by, Alan Parker), (Bugsy Malone, directed_by, Alan Parker), (Bugsy Malone, release_year, 1976)}	Input Question and Reference Answers	Question: What is the most current year the team who won the 2010 World Series has won another World Series? Reference answers: A. 2010 World Series (correct probability: 0.487) {relevant facts: (San Francisco Giants, time.participant.event, 2010 World Series), (San Francisco Giants, sports.sports_team.championships, 2010 World Series)} B. 2014 World Series (correct probability: 0.26) {relevant facts: (San Francisco Giants, sports.sports_team.championships, 2010 World Series), (San Francisco Giants, sports.sports_team.championships, 2014 World Series)} C. 2012 World Series (correct probability: 0.253) {relevant facts: (San Francisco Giants, time.participant.event, 2010 World Series), (San Francisco Giants, sports.sports_team.championships, 2012 World Series)}
Output by DualR-Llama2-13B	Based on the reference answers and my own knowledge, I can answer the question. The correct answer is A. 1989. Here’s why: The reference answers provide three relevant facts: 1. (Birdy, written_by, William Wharton) 2. (Dad, written_by, William Wharton) 3. (Dad, release_year, 1989) Based on these facts, we can infer that the film “Dad” was written by William Wharton and released in 1989. Therefore, the correct answer is A. 1989.	Output by DualR-Llama2-13B	Based on the information provided, the most current year the team who won the 2010 World Series has won another World Series is: B. 2014 World Series. The relevant facts provided in the reference answer indicates that San Francisco Giants won 2010 World Series and 2014 World Series. Therefore, the answer is B. 2014 World Series.

(a) One example from MetaQA-3hop.

(b) One example from CWQ.

Figure 3: Examples of accurate and faithful output by DualR-Llama2-13B.

5. Conclusion

In this paper, we introduce Dual-Reasoning (DualR), a novel framework following dual-process theory, which combines GNN-based structured, deliberate reasoning of “System 2” with LLM-based intuitive reasoning of “System 1” in KGQA task. DualR designs an LLM-empowered GNN module to explicitly reason on KGs, extracting high-quality reasoning chains, which are then refined into a knowledge-enhanced multiple-choice prompt, guiding the LLM to reason for final answer. With the collaboration of GNN and LLM, this two-tier reasoning process harnesses the precision graph learning of GNN and the language understanding prowess of LLM. Extensive experiments demonstrate the superiority of DualR, achieving accurate, efficient and faithful reasoning over KGs.